THREE EQUIVALENT DESCRIPTIONS OF BOUNDED INJECTIVE AMPLITUDE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Injective amplitude of a complex, defined through the existence of a bounded complex of injectives quasi-isomorphic to it. defined earlier in the same section, source lines 17480-17528, before the counted statement. Supplies the property appearing in the first condition.
- Computation of Ext groups by a complex of injectives, and the exactness of the functor Hom from the ring. used at proof lines 17542-17560. Gives the vanishing statements in the second and third conditions.
- Existence of a right resolution of a bounded below complex by injective modules. the author cites Derived Categories, Lemma derived-lemma-subcategory-right-resolution at proof lines 17570-17572. Produces the injective resolution used to compare with the injective amplitude condition.

Source reference. Original lemma 0A5T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12821–12821. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 05T6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7470–7470. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AVF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13683–13683. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let K be an object of $D(R)$. Let $a, b \in \mathbf{Z}$. The following are equivalent*

(1) *K has injective-amplitude in $[a, b]$,*
(2) *$\mathrm{Ext}^i_R(N, K) = 0$ for all R-modules N and all $i \notin [a, b]$,*
(3) *$\mathrm{Ext}^i(R/I, K) = 0$ for all ideals $I \subset R$ and all $i \notin [a, b]$.*

Proof. Assume (1). We may assume K is the complex

$$\ldots \to 0 \to I^a \to I^{a+1} \to \ldots \to I^{b-1} \to I^b \to 0 \to \ldots$$

where I^i is an injective R-module for all $i \in \mathbf{Z}$. In this case we can compute the ext groups by the complex

$$\ldots \to 0 \to \mathrm{Hom}_R(N, I^a) \to \ldots \to \mathrm{Hom}_R(N, I^b) \to 0 \to \ldots$$

and we obtain (2). It is clear that (2) implies (3).

Assume (3) holds. Choose a nonzero map $R \to H^n(K)$. Since $\mathrm{Hom}_R(R, -)$ is an exact functor, we see that $\mathrm{Ext}^n_R(R, K) = \mathrm{Hom}_R(R, H^n(K)) = H^n(K)$. We conclude that $H^n(K)$ is zero for $n \notin [a, b]$. In particular, K is bounded below and we can choose a quasi-isomorphism

$$K \to I^\bullet$$

with I^i injective for all $i \in \mathbf{Z}$ and $I^i = 0$ for $i < a$. See Derived Categories, Lemma 05T6. Let $J = \mathrm{Ker}(I^b \to I^{b+1})$. Then K is quasi-isomorphic to the complex

$$\ldots \to 0 \to I^a \to \ldots \to I^{b-1} \to J \to 0 \to \ldots$$

Denote $K' = (I^a \to \ldots \to I^{b-1})$ the corresponding object of $D(R)$. We obtain a distinguished triangle

$$J[-b] \to K \to K' \to J[1-b]$$

in $D(R)$. Thus for every ideal $I \subset R$ an exact sequence

$$\mathrm{Ext}^b(R/I, K') \to \mathrm{Ext}^1(R/I, J) \to \mathrm{Ext}^{1+b}(R/I, K)$$

By assumption the term on the right vanishes. By the implication (1) $\Rightarrow$ (2) the term on the left vanishes. Thus J is an injective R-module by Lemma 0AVF. $\square$